Calculus

1. L'Hôpital's Rule
2. Extreme Values
3. Concavity and Inflections; Sketching the Graph of a Function
4. Extreme-Value Problems
5. Linear Approximations
6. Taylor Polynomials
7. Integration by Parts
8. Integrals of Rational Functions
9. Inverse Substitutions
10. Other Methods for Evaluating Integrals
11. Improper Integrals

L'Hôpital's Rule

Table 1: Typical Intermediate Forms of Limit

Form	Example

We will focus on $\frac{0}{0}, \frac{\infty}{\infty}$; Other forms can sometimes be transformed to these two

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1^∞	$\lim\limits_{x\to \infty} \left(1 + \frac{1}{x}\right)^x$

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Let f and g be real-valued and differentiable with

$$g' \neq 0 \tag{Denominator$' \neq 0$}$$

on (a, b), $a < b$, $a, b \in \mathbb{R}^\sharp$. Suppose

$$\left(\frac{0}{0}\right)$$

or

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If

$$\lim_{x \downarrow a} \frac{f'(x)}{g'(x)} = L \in \mathbb{R}^\sharp$$

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The result holds if $x \downarrow a$ is replaced by $x \uparrow b$ and/or ∞ is replaced by $-\infty$.

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L'Hôpital's Rule: Example II

Find

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L'Hôpital's Rule: Example III

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L'Hôpital's Rule: Example III (Cont'd)

Using L'Hôpital's rule again with

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Using L'Hôpital's rule again with

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Checking the conditions (Denominator' $\neq 0$) and ($\frac{0}{0}$):

(Denominator' $\neq 0$) $\left(1-\frac{1}{x}+\ln x\right)' = \frac{1+x}{x^2} \neq 0$ on $(1,b)$, $b>1$.

($\frac{0}{0}$) $\lim_{x\to 1+} \frac{\ln x}{1-\frac{1}{x}+\ln x}$ is of $\frac{0}{0}$ form: $\lim_{x\to 1+} \ln x = \lim_{x\to 1+} 1-\frac{1}{x}+\ln x = 0$

Hence

$$\lim_{x\to 1+} \frac{\ln x}{1-\frac{1}{x}+\ln x} = \lim_{x\to 1+} \frac{(\ln x)'}{\left(1-\frac{1}{x}+\ln x\right)'} = \lim_{x\to 1+} \frac{\frac{1}{x}}{\frac{1+x}{x^2}} = \frac{1}{2}$$

So

$$\lim_{x\to 1+} \left(\frac{x}{x-1}-\frac{1}{\ln x}\right) = \lim_{x\to 1+} \frac{x\ln x-(x-1)}{(x-1)\ln x} = \lim_{x\to 1+} \frac{\ln x}{1-\frac{1}{x}+\ln x} = \frac{1}{2}$$

L'Hôpital's Rule: Example IV

Find

$$\lim_{x \to 0} \left(\frac{\sin x}{x} \right)^{\frac{1}{x^2}}$$

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We consider the logarithm

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Hence

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Using L'Hôpital's rule again with

$$\lim_{x \to 0} \frac{x \cos x - \sin x}{2x^2 \sin x}$$

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Checking the conditions (Denominator' $\neq 0$) and $(\frac{0}{0})$:

(Denominator' $\neq 0$) $(2x^2 \sin x)' = 2x^2 \cos x + 4x \sin x \neq 0$ on $(0, b)$, $b > 0$ is small.

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L'Hôpital's Rule: Example IV (Cont'd)

Using L'Hôpital's rule again with

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Hence

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Using L'Hôpital's rule again with

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So

$$\begin{aligned}\lim_{x\to 0} \left(\frac{\sin x}{x}\right)^{\frac{1}{x^2}} &= \exp\left\{\lim_{x\to 0} \frac{1}{x^2} \ln \frac{\sin x}{x}\right\} \\ &= \exp\left\{\lim_{x\to 0} \frac{x\cos x - \sin x}{2x^2 \sin x}\right\} = \exp\left\{\lim_{x\to 0} \frac{-\sin x}{2x\cos x + 4\sin x}\right\} = e^{-\frac{1}{6}}\end{aligned}$$

(4.3.29)

$$\lim_{x\to 0} (\cos 2x)^{\frac{1}{x^2}}$$

(4.3.30)

$$\lim_{x\to 0+} \frac{\csc x}{\ln x}$$

(4.3.31)

$$\lim_{x\to 0} \frac{\ln \sin \pi x}{\csc \pi x}$$

(4.3.32)

$$\lim_{x\to 0} (1 + \tan x)^{\frac{1}{x}}$$

(4.3.33) For twice differentiable f, find

$$\lim_{h\to 0} \frac{f(x+h) - 2f(x) + f(x-h)}{h^2}$$

(4.3.34) If f has continuous third derivative, find

$$\lim_{h\to 0} \frac{f(x+3h) - 3f(x+h) + 3f(x-h) - f(x-3h)}{h^3}$$

Extreme Values

Let $f : I \mapsto \mathbb{R}$.

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Examples:

$f(x) = \sin x$ on $I = \mathbb{R}$

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Examples:

$f(x) = \sin x$ on $I = \mathbb{R}$ has absolute maximum 1 and absolute minimum -1

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Examples:

$f(x) = \sin x$ on $I = \mathbb{R}$ has absolute maximum 1 and absolute minimum -1

$f(x) = \frac{1}{x}$ on $I = \mathbb{R}$

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Examples:

$f(x) = \sin x$ on $I = \mathbb{R}$ has absolute maximum 1 and absolute minimum -1

$f(x) = \frac{1}{x}$ on $I = \mathbb{R}$ has no absolute maximum and minimum

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The maximum is $f(-1) = 7$, the minimum is $f(2) = -20$.

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The minimum value is $f(2) = 4$.

Concavity and Inflections; Sketching the Graph of a Function

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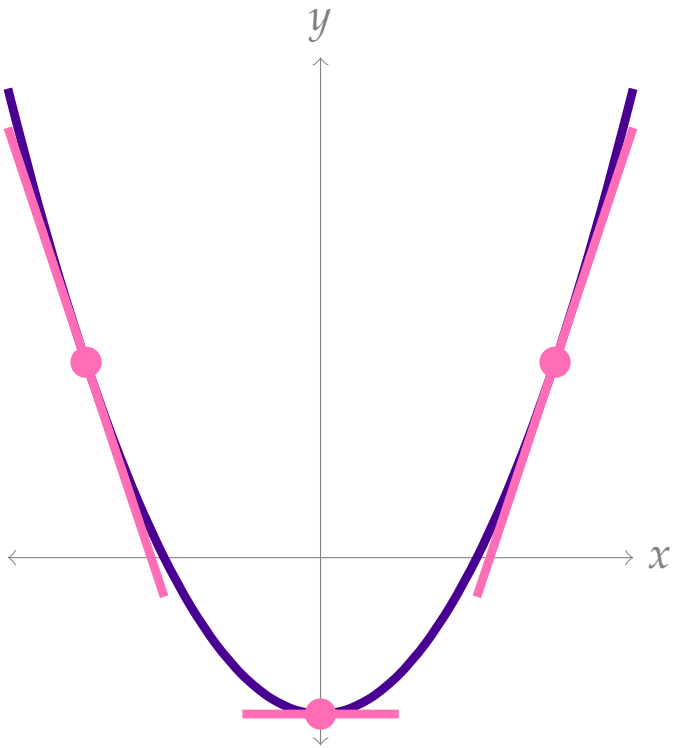
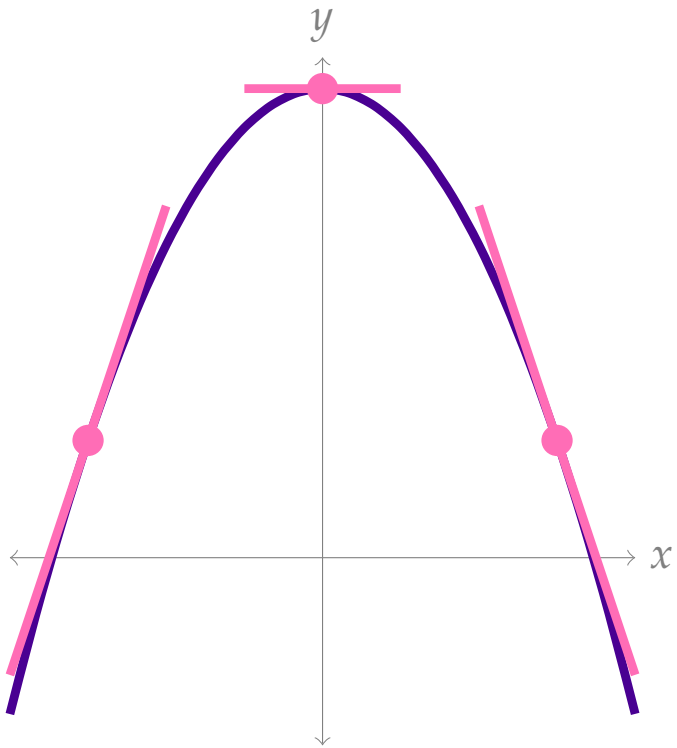


Slopes are increasing

$f''(x) > 0$

"concave up"

tangent line below curve



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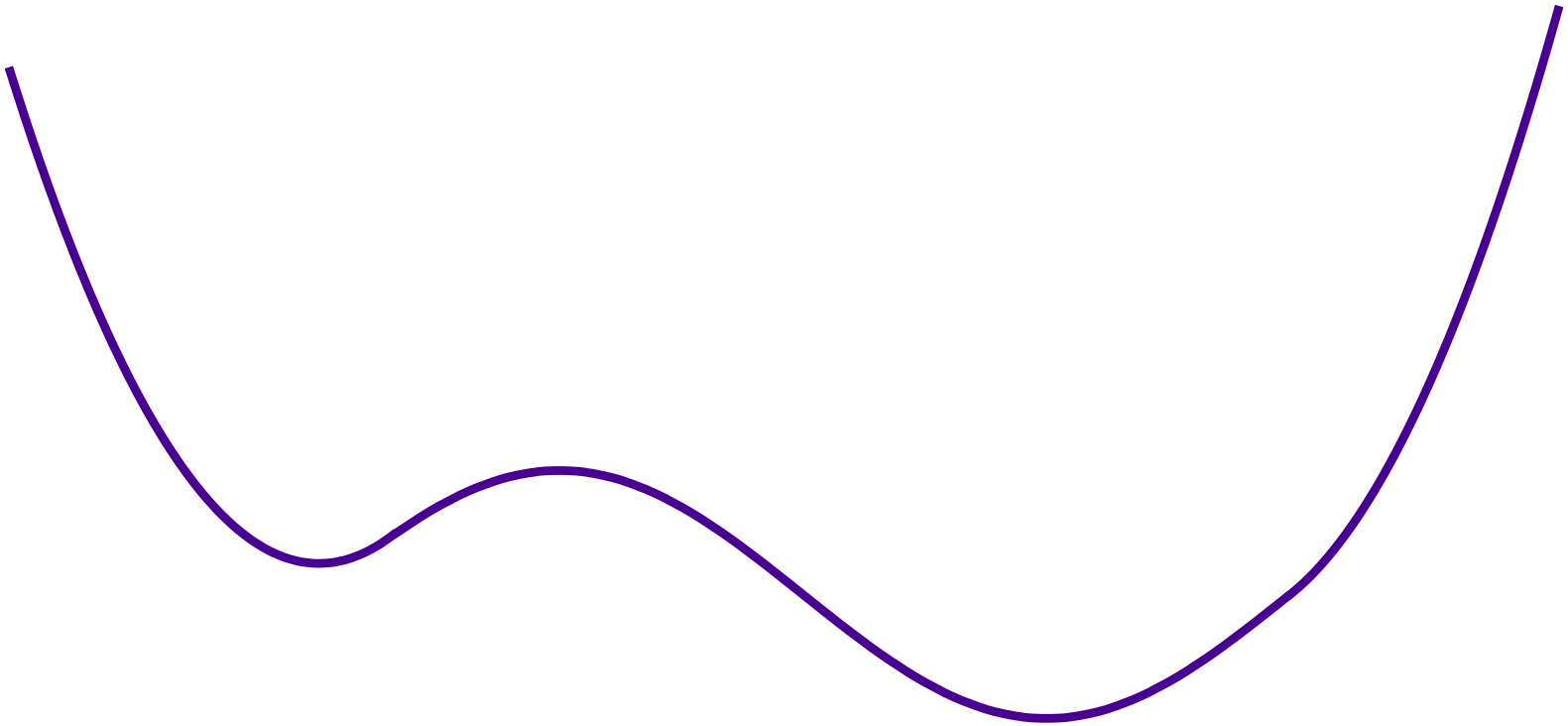

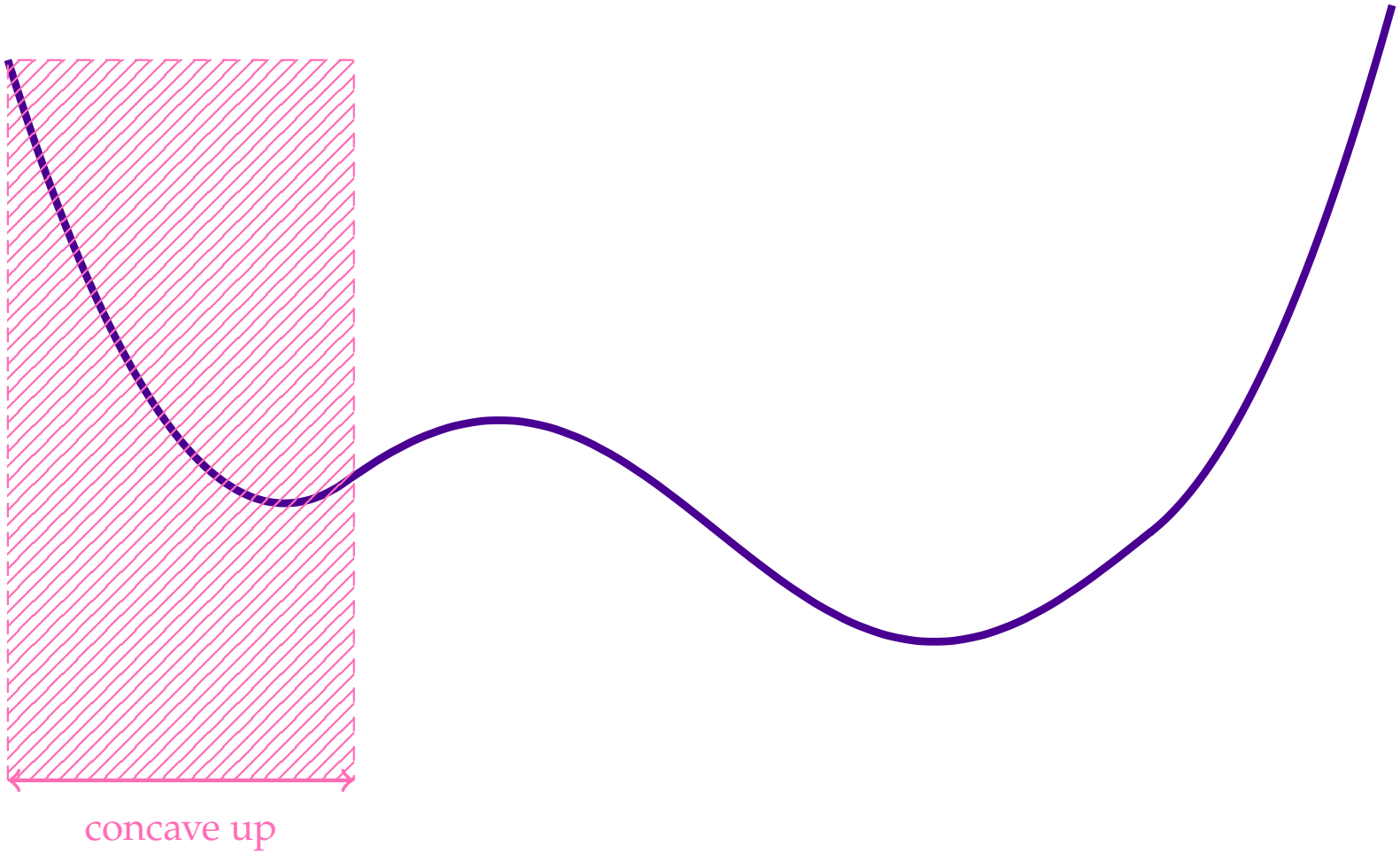
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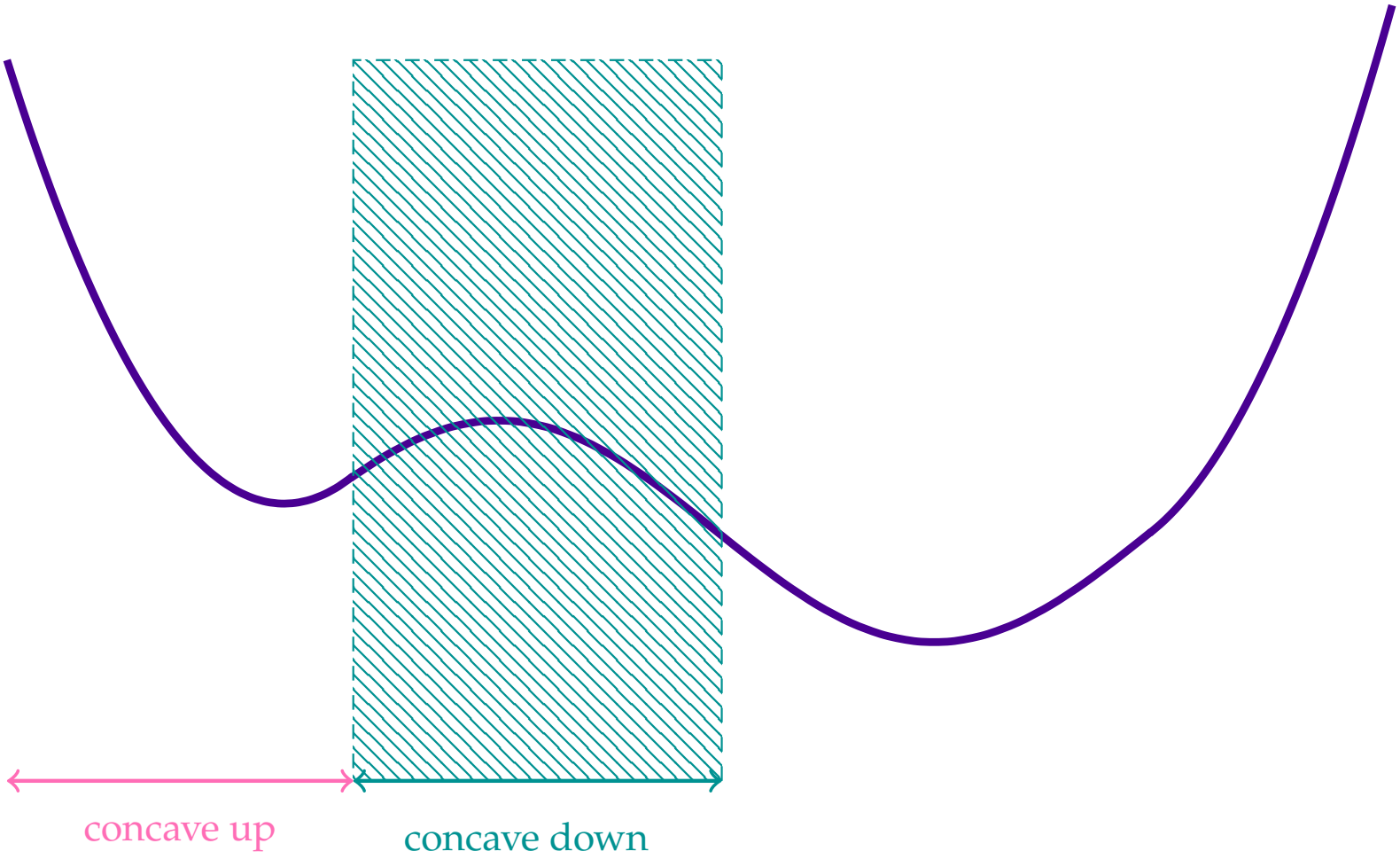
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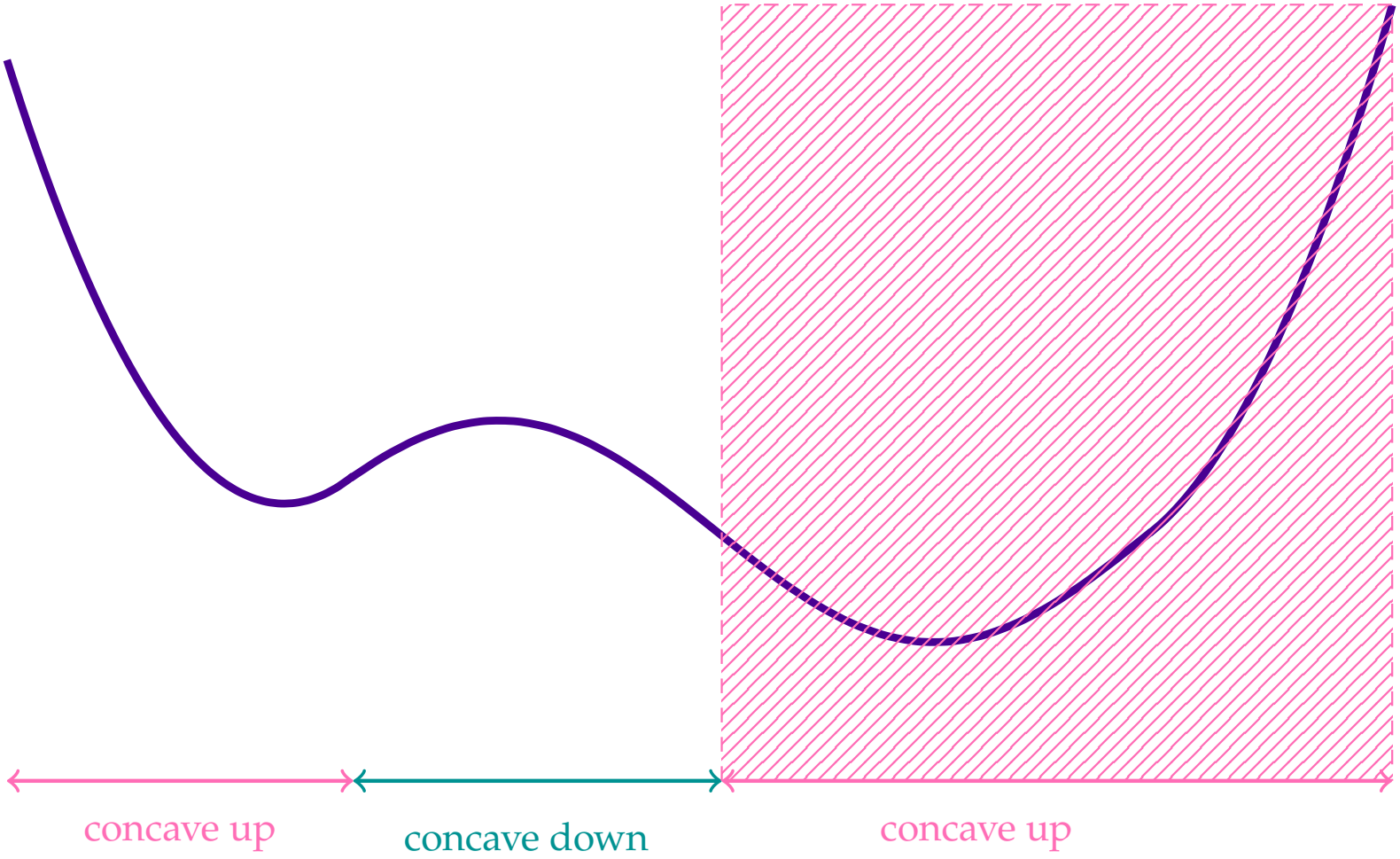
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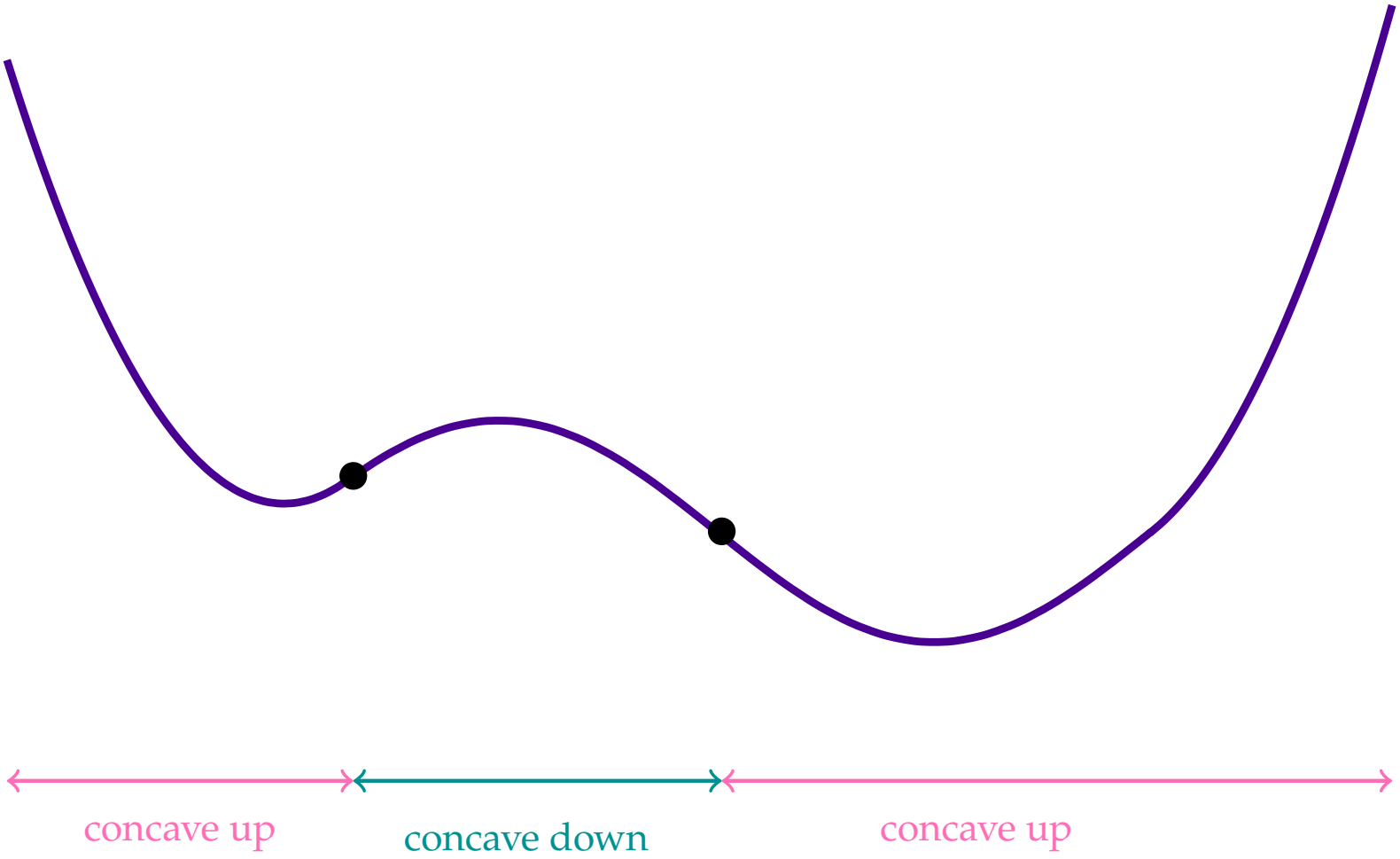
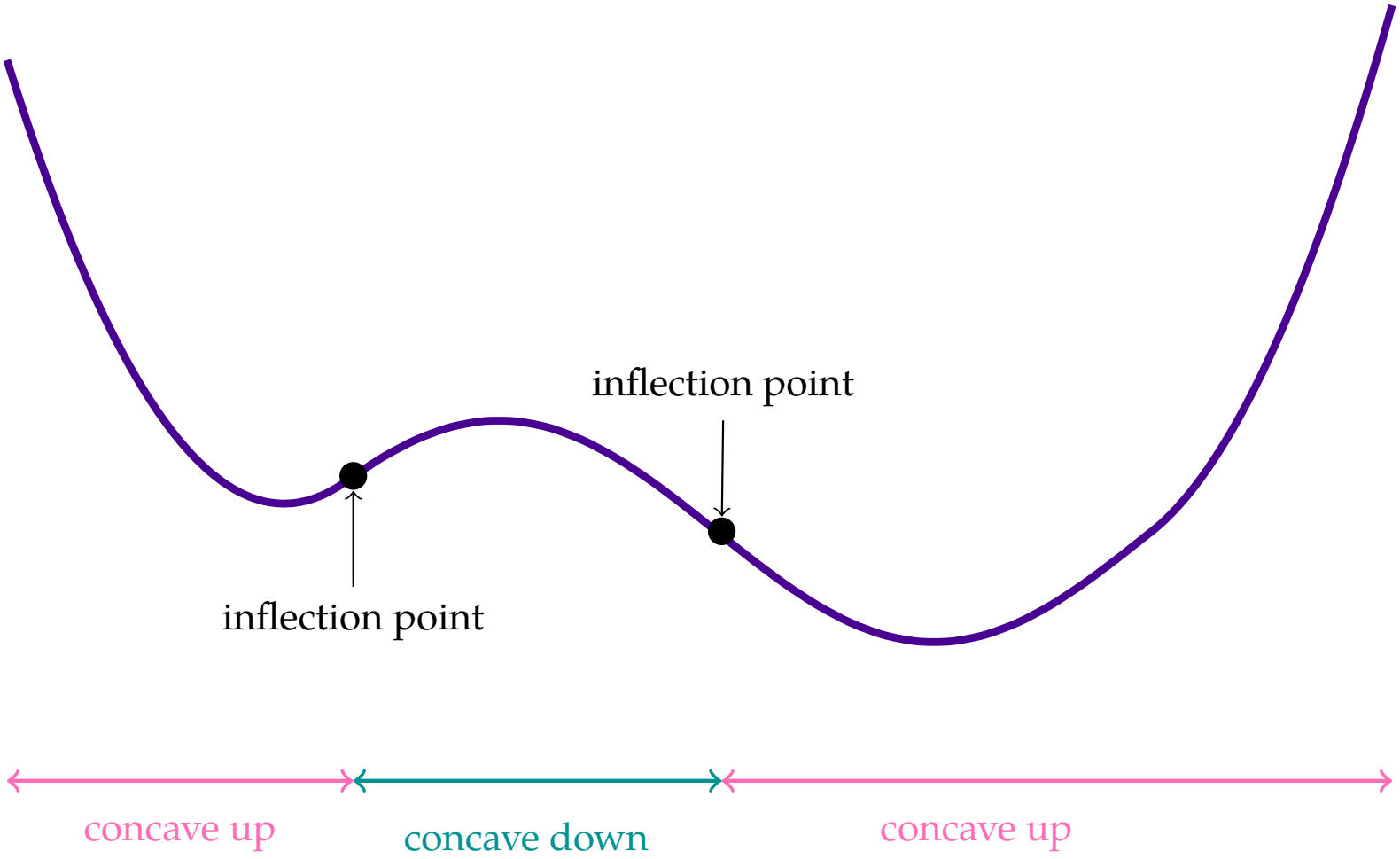
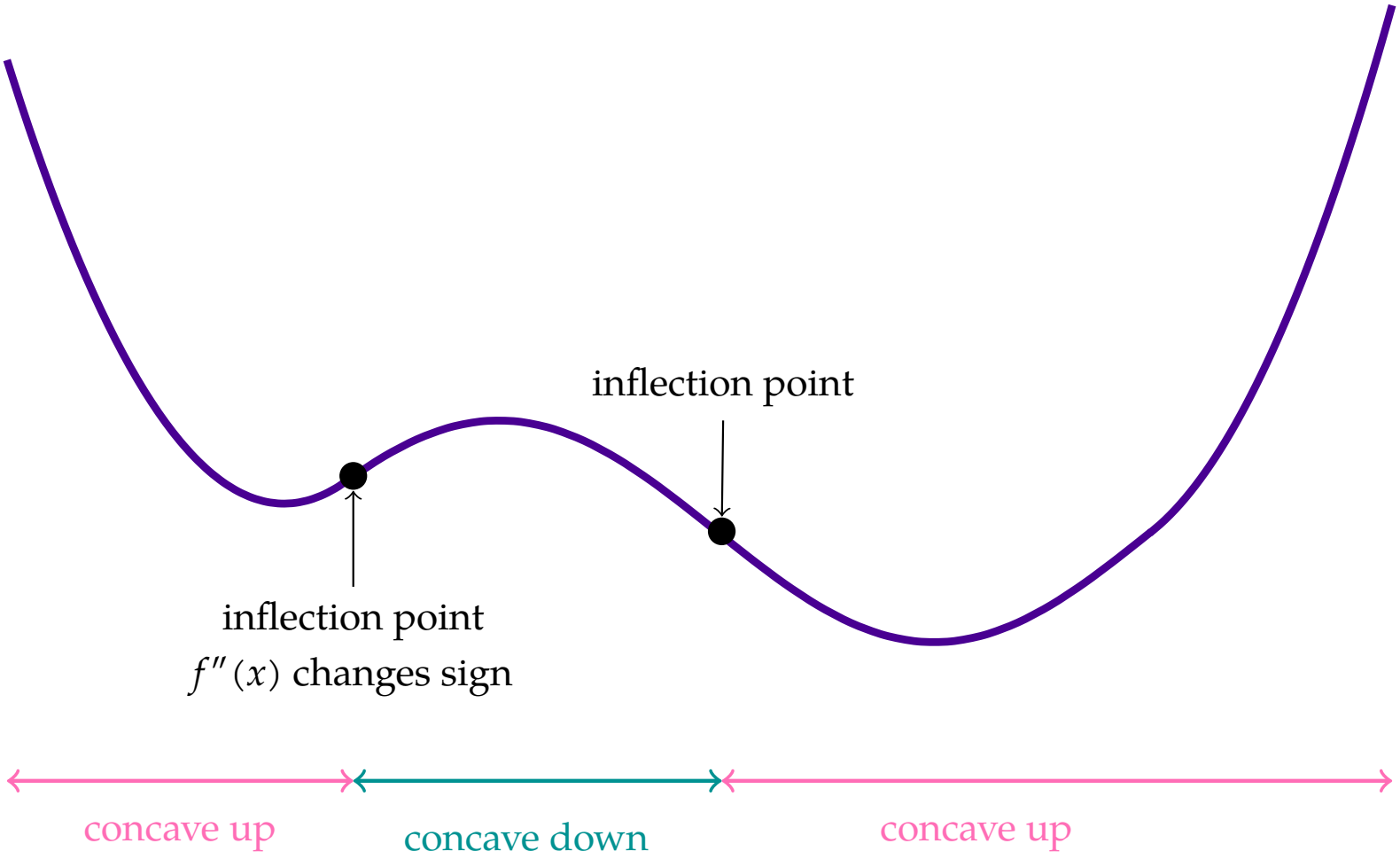
concave down
concave up


concave up
concave down
concave up


inflection point
inflection point
concave up
concave down
concave up


inflection point
inflection point
$f''(x)$ changes sign
concave up
concave down
concave up

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\ (a, b)$ containing x_0 s.t.

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\ (a, b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\ (a, b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b),

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\ (a, b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\ (a, b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\ (a, b)$ containing x_0 s.t.

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\ (a, b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\ (a, b)$ containing x_0 s.t. $f'(x) < 0$ on (a, x_0) and

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\ (a, b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\ (a, b)$ containing x_0 s.t. $f'(x) < 0$ on (a, x_0) and $f'(x) > 0$ on (x_0, b),

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\ (a, b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\ (a, b)$ containing x_0 s.t. $f'(x) < 0$ on (a, x_0) and $f'(x) > 0$ on (x_0, b), then f has a local minimum value at x_0.

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\ (a, b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\ (a, b)$ containing x_0 s.t. $f'(x) < 0$ on (a, x_0) and $f'(x) > 0$ on (x_0, b), then f has a local minimum value at x_0.

The First Derivative Test II: Let a be the left endpoint of the domain of f and f is right continuous at a.

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) < 0$ on (a, x_0) and $f'(x) > 0$ on (x_0, b), then f has a local minimum value at x_0.

The First Derivative Test II: Let a be the left endpoint of the domain of f and f is right continuous at a.

1. If $f'(x) > 0$ on $(a, a+h)$, $h > 0$,

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\ (a,b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\ (a,b)$ containing x_0 s.t. $f'(x) < 0$ on (a, x_0) and $f'(x) > 0$ on (x_0, b), then f has a local minimum value at x_0.

The First Derivative Test II: Let a be the left endpoint of the domain of f and f is right continuous at a.

1. If $f'(x) > 0$ on $(a, a+h), h > 0$, then f has a local minimum value at a.

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) < 0$ on (a, x_0) and $f'(x) > 0$ on (x_0, b), then f has a local minimum value at x_0.

The First Derivative Test II: Let a be the left endpoint of the domain of f and f is right continuous at a.

1. If $f'(x) > 0$ on $(a, a+h), h > 0$, then f has a local minimum value at a.
2. If $f'(x) < 0$ on $(a, a+h), h > 0$,

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) < 0$ on (a, x_0) and $f'(x) > 0$ on (x_0, b), then f has a local minimum value at x_0.

The First Derivative Test II: Let a be the left endpoint of the domain of f and f is right continuous at a.

1. If $f'(x) > 0$ on $(a, a+h), h > 0$, then f has a local minimum value at a.
2. If $f'(x) < 0$ on $(a, a+h), h > 0$, then f has a local maximum value at a.

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) < 0$ on (a, x_0) and $f'(x) > 0$ on (x_0, b), then f has a local minimum value at x_0.

The First Derivative Test II: Let a be the left endpoint of the domain of f and f is right continuous at a.

1. If $f'(x) > 0$ on $(a, a+h), h > 0$, then f has a local minimum value at a.
2. If $f'(x) < 0$ on $(a, a+h), h > 0$, then f has a local maximum value at a.

The First Derivative Test III: Let b be the right endpoint of the domain of f and f is left continuous at b.

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) < 0$ on (a, x_0) and $f'(x) > 0$ on (x_0, b), then f has a local minimum value at x_0.

The First Derivative Test II: Let a be the left endpoint of the domain of f and f is right continuous at a.

1. If $f'(x) > 0$ on $(a, a+h), h > 0$, then f has a local minimum value at a.
2. If $f'(x) < 0$ on $(a, a+h), h > 0$, then f has a local maximum value at a.

The First Derivative Test III: Let b be the right endpoint of the domain of f and f is left continuous at b.

1. If $f'(x) > 0$ on $(b-h, b), h > 0$,

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) < 0$ on (a, x_0) and $f'(x) > 0$ on (x_0, b), then f has a local minimum value at x_0.

The First Derivative Test II: Let a be the left endpoint of the domain of f and f is right continuous at a.

1. If $f'(x) > 0$ on $(a, a+h), h > 0$, then f has a local minimum value at a.
2. If $f'(x) < 0$ on $(a, a+h), h > 0$, then f has a local maximum value at a.

The First Derivative Test III: Let b be the right endpoint of the domain of f and f is left continuous at b.

1. If $f'(x) > 0$ on $(b-h, b), h > 0$, then f has a local maximum value at b.

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) < 0$ on (a, x_0) and $f'(x) > 0$ on (x_0, b), then f has a local minimum value at x_0.

The First Derivative Test II: Let a be the left endpoint of the domain of f and f is right continuous at a.

1. If $f'(x) > 0$ on $(a, a+h), h > 0$, then f has a local minimum value at a.
2. If $f'(x) < 0$ on $(a, a+h), h > 0$, then f has a local maximum value at a.

The First Derivative Test III: Let b be the right endpoint of the domain of f and f is left continuous at b.

1. If $f'(x) > 0$ on $(b-h, b), h > 0$, then f has a local maximum value at b.
2. If $f'(x) < 0$ on $(b-h, b), h > 0$,

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) < 0$ on (a, x_0) and $f'(x) > 0$ on (x_0, b), then f has a local minimum value at x_0.

The First Derivative Test II: Let a be the left endpoint of the domain of f and f is right continuous at a.

1. If $f'(x) > 0$ on $(a, a+h), h > 0$, then f has a local minimum value at a.
2. If $f'(x) < 0$ on $(a, a+h), h > 0$, then f has a local maximum value at a.

The First Derivative Test III: Let b be the right endpoint of the domain of f and f is left continuous at b.

1. If $f'(x) > 0$ on $(b-h, b), h > 0$, then f has a local maximum value at b.
2. If $f'(x) < 0$ on $(b-h, b), h > 0$, then f has a local minimum value at b.

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) < 0$ on (a, x_0) and $f'(x) > 0$ on (x_0, b), then f has a local minimum value at x_0.

The First Derivative Test II: Let a be the left endpoint of the domain of f and f is right continuous at a.

1. If $f'(x) > 0$ on $(a, a+h), h > 0$, then f has a local minimum value at a.
2. If $f'(x) < 0$ on $(a, a+h), h > 0$, then f has a local maximum value at a.

The First Derivative Test III: Let b be the right endpoint of the domain of f and f is left continuous at b.

1. If $f'(x) > 0$ on $(b-h, b), h > 0$, then f has a local maximum value at b.
2. If $f'(x) < 0$ on $(b-h, b), h > 0$, then f has a local minimum value at b.

The Second Derivative Test

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) < 0$ on (a, x_0) and $f'(x) > 0$ on (x_0, b), then f has a local minimum value at x_0.

The First Derivative Test II: Let a be the left endpoint of the domain of f and f is right continuous at a.

1. If $f'(x) > 0$ on $(a, a+h), h > 0$, then f has a local minimum value at a.
2. If $f'(x) < 0$ on $(a, a+h), h > 0$, then f has a local maximum value at a.

The First Derivative Test III: Let b be the right endpoint of the domain of f and f is left continuous at b.

1. If $f'(x) > 0$ on $(b-h, b), h > 0$, then f has a local maximum value at b.
2. If $f'(x) < 0$ on $(b-h, b), h > 0$, then f has a local minimum value at b.

The Second Derivative Test

If $f'(x_0) = 0$ and $f''(x_0) < 0$,

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) < 0$ on (a, x_0) and $f'(x) > 0$ on (x_0, b), then f has a local minimum value at x_0.

The First Derivative Test II: Let a be the left endpoint of the domain of f and f is right continuous at a.

1. If $f'(x) > 0$ on $(a, a+h), h > 0$, then f has a local minimum value at a.
2. If $f'(x) < 0$ on $(a, a+h), h > 0$, then f has a local maximum value at a.

The First Derivative Test III: Let b be the right endpoint of the domain of f and f is left continuous at b.

1. If $f'(x) > 0$ on $(b-h, b), h > 0$, then f has a local maximum value at b.
2. If $f'(x) < 0$ on $(b-h, b), h > 0$, then f has a local minimum value at b.

The Second Derivative Test

If $f'(x_0) = 0$ and $f''(x_0) < 0$, then f has a local maximum value at x_0.

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) < 0$ on (a, x_0) and $f'(x) > 0$ on (x_0, b), then f has a local minimum value at x_0.

The First Derivative Test II: Let a be the left endpoint of the domain of f and f is right continuous at a.

1. If $f'(x) > 0$ on $(a, a+h), h > 0$, then f has a local minimum value at a.
2. If $f'(x) < 0$ on $(a, a+h), h > 0$, then f has a local maximum value at a.

The First Derivative Test III: Let b be the right endpoint of the domain of f and f is left continuous at b.

1. If $f'(x) > 0$ on $(b-h, b), h > 0$, then f has a local maximum value at b.
2. If $f'(x) < 0$ on $(b-h, b), h > 0$, then f has a local minimum value at b.

The Second Derivative Test

If $f'(x_0) = 0$ and $f''(x_0) < 0$, then f has a local maximum value at x_0.

If $f'(x_0) = 0$ and $f''(x_0) > 0$,

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) < 0$ on (a, x_0) and $f'(x) > 0$ on (x_0, b), then f has a local minimum value at x_0.

The First Derivative Test II: Let a be the left endpoint of the domain of f and f is right continuous at a.

1. If $f'(x) > 0$ on $(a, a+h), h > 0$, then f has a local minimum value at a.
2. If $f'(x) < 0$ on $(a, a+h), h > 0$, then f has a local maximum value at a.

The First Derivative Test III: Let b be the right endpoint of the domain of f and f is left continuous at b.

1. If $f'(x) > 0$ on $(b-h, b), h > 0$, then f has a local maximum value at b.
2. If $f'(x) < 0$ on $(b-h, b), h > 0$, then f has a local minimum value at b.

The Second Derivative Test

If $f'(x_0) = 0$ and $f''(x_0) < 0$, then f has a local maximum value at x_0.

If $f'(x_0) = 0$ and $f''(x_0) > 0$, then f has a local minimum value at x_0.

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) < 0$ on (a, x_0) and $f'(x) > 0$ on (x_0, b), then f has a local minimum value at x_0.

The First Derivative Test II: Let a be the left endpoint of the domain of f and f is right continuous at a.

1. If $f'(x) > 0$ on $(a, a+h), h > 0$, then f has a local minimum value at a.
2. If $f'(x) < 0$ on $(a, a+h), h > 0$, then f has a local maximum value at a.

The First Derivative Test III: Let b be the right endpoint of the domain of f and f is left continuous at b.

1. If $f'(x) > 0$ on $(b-h, b), h > 0$, then f has a local maximum value at b.
2. If $f'(x) < 0$ on $(b-h, b), h > 0$, then f has a local minimum value at b.

The Second Derivative Test

If $f'(x_0) = 0$ and $f''(x_0) < 0$, then f has a local maximum value at x_0.

If $f'(x_0) = 0$ and $f''(x_0) > 0$, then f has a local minimum value at x_0.

If $f'(x_0) = 0$ and $f''(x_0) = 0$,

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) < 0$ on (a, x_0) and $f'(x) > 0$ on (x_0, b), then f has a local minimum value at x_0.

The First Derivative Test II: Let a be the left endpoint of the domain of f and f is right continuous at a.

1. If $f'(x) > 0$ on $(a, a+h), h > 0$, then f has a local minimum value at a.
2. If $f'(x) < 0$ on $(a, a+h), h > 0$, then f has a local maximum value at a.

The First Derivative Test III: Let b be the right endpoint of the domain of f and f is left continuous at b.

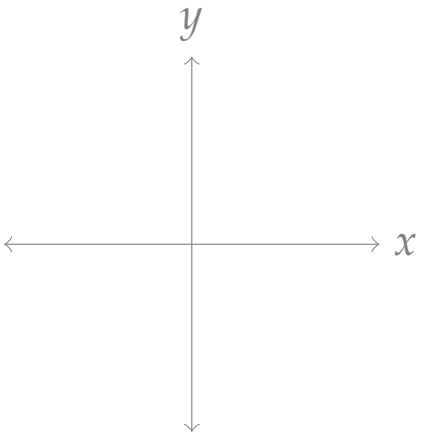
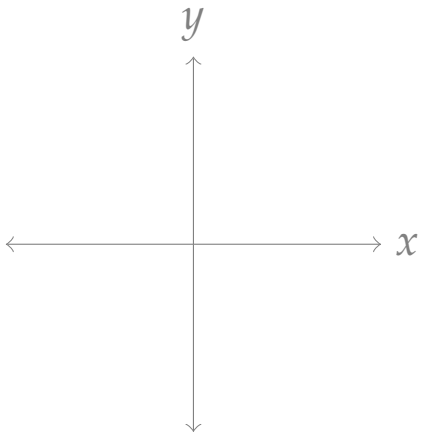
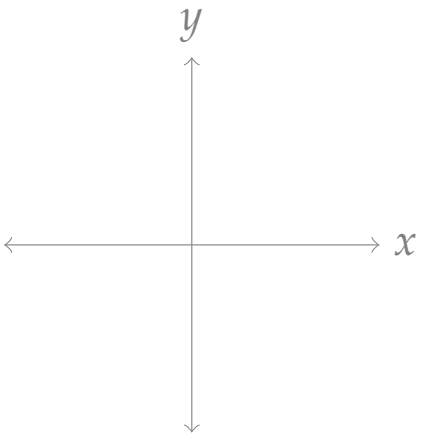
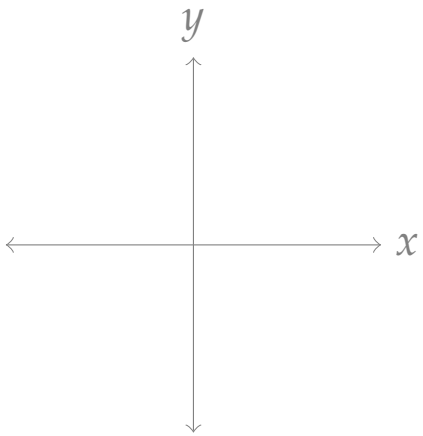
1. If $f'(x) > 0$ on $(b-h, b), h > 0$, then f has a local maximum value at b.
2. If $f'(x) < 0$ on $(b-h, b), h > 0$, then f has a local minimum value at b.

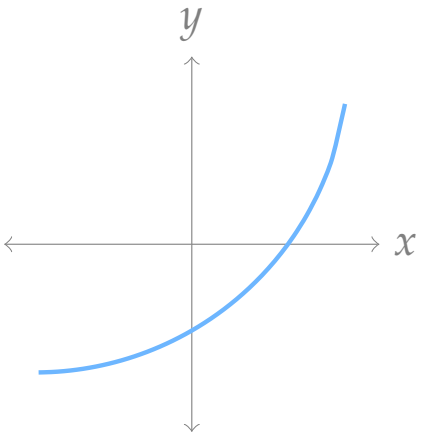
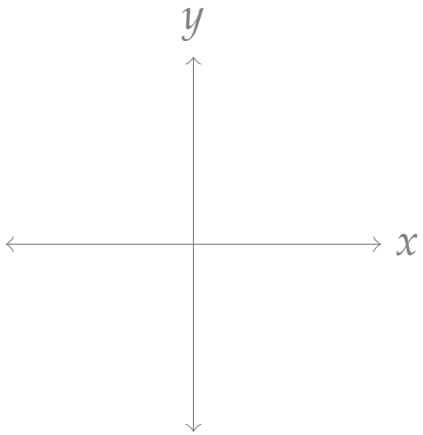
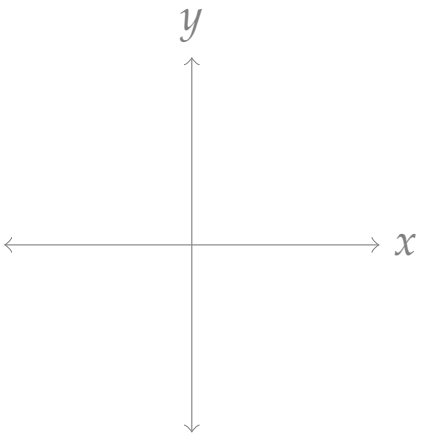
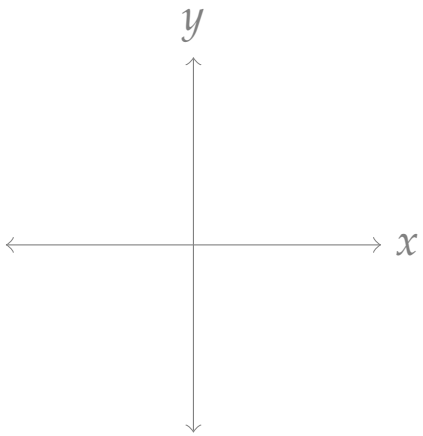
The Second Derivative Test

If $f'(x_0) = 0$ and $f''(x_0) < 0$, then f has a local maximum value at x_0.

If $f'(x_0) = 0$ and $f''(x_0) > 0$, then f has a local minimum value at x_0.

If $f'(x_0) = 0$ and $f''(x_0) = 0$, then no conclusion can be drawn.

	concave up ($f'' > 0$)	concave down ($f'' < 0$)
increasing ($f' > 0$)		
decreasing ($f' < 0$)		

	concave up ($f'' > 0$)	concave down ($f'' < 0$)
increasing ($f' > 0$)		
decreasing ($f' < 0$)		

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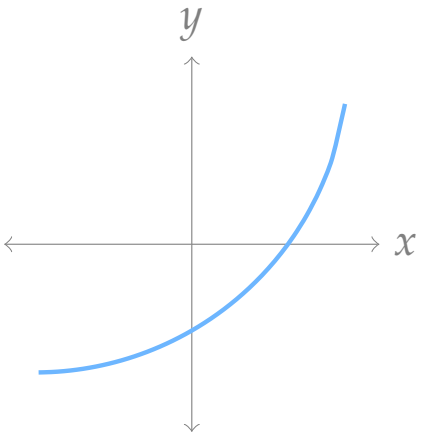
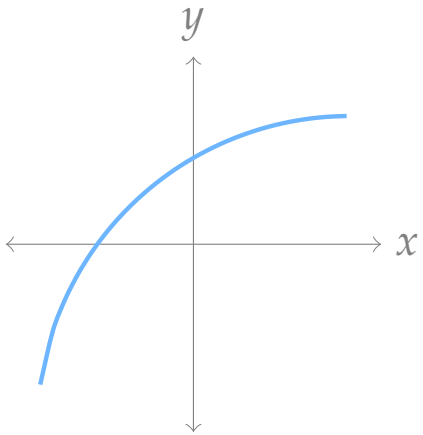
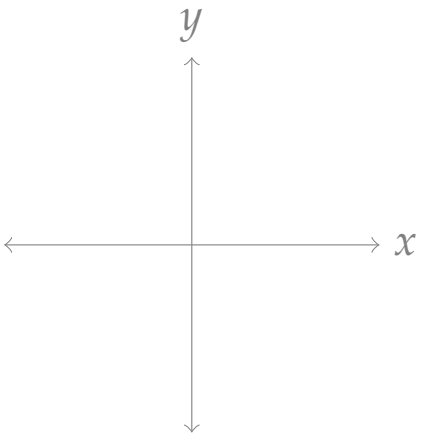
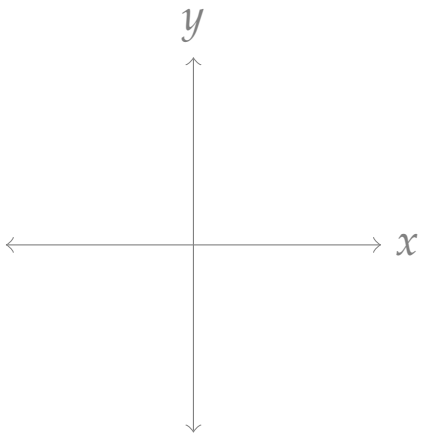
increasing ($f' > 0$)		
decreasing ($f' < 0$)		

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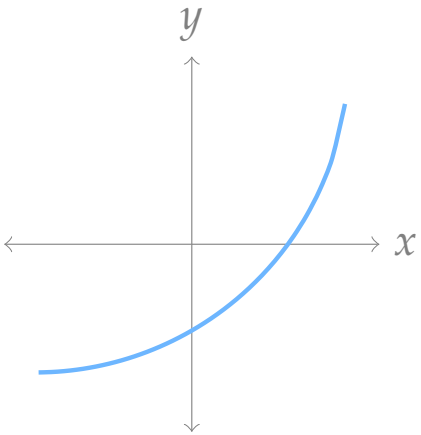
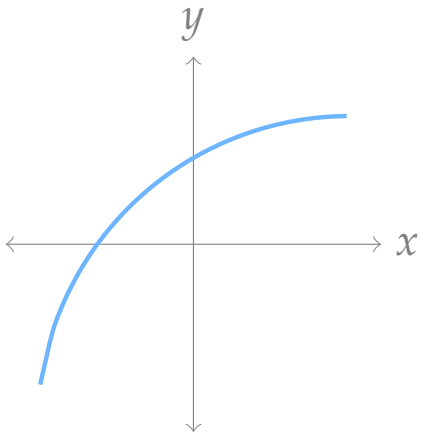
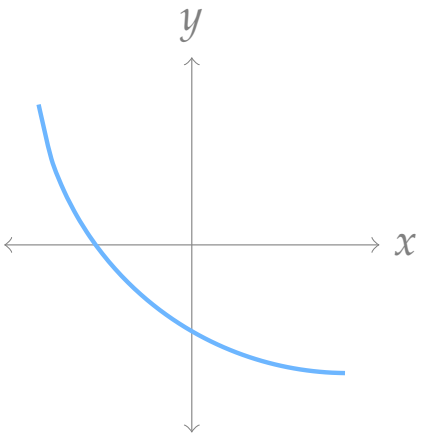
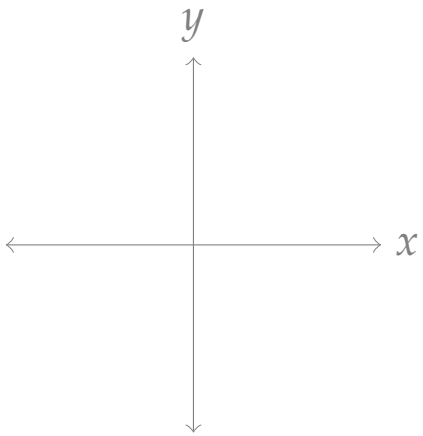
increasing ($f' > 0$)		
decreasing ($f' < 0$)		

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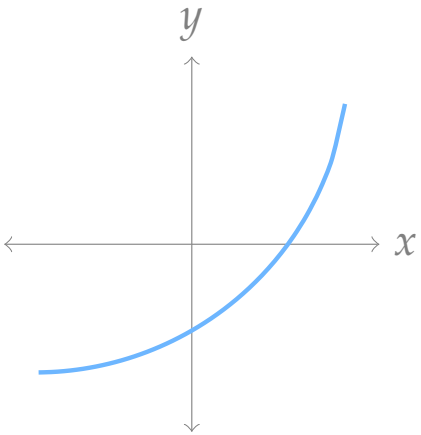
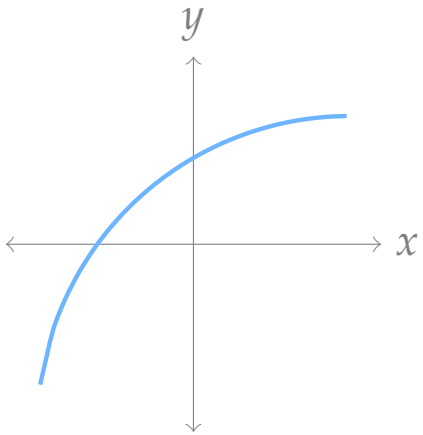
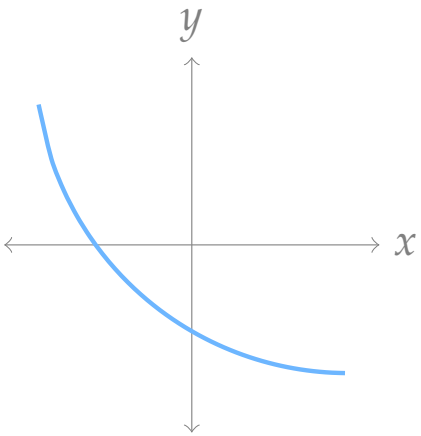
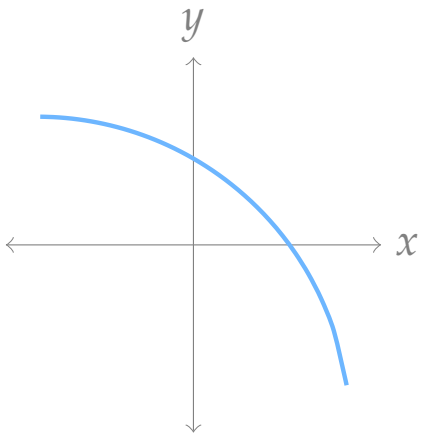
increasing ($f' > 0$)		
decreasing ($f' < 0$)		

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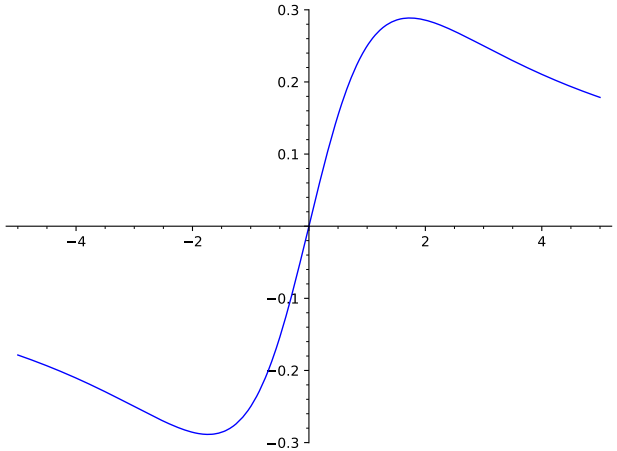

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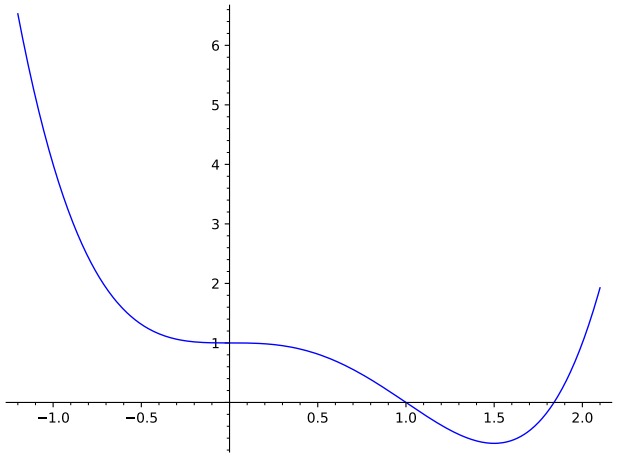

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Determine the intervals of increase and decrease, the local extreme values, the concavity and the graph of $f(x) = \frac{x^3}{x^2-1}$.

$f'(x) = \frac{x^2(x^2-3)}{(x^2-1)^2} = \frac{x^2(x-\sqrt{3})(x+\sqrt{3})}{(x^2-1)^2}$, $f''(x) = \frac{2x(x^2+3)}{(x^2-1)^3}$.

List the critical, singular, and inflection points and decide the sign of f', f'' in each interval:

	$+$		$-$		$-$		$-$		$-$		$+$	f'
$-$		$-\sqrt{3}$	$-$	-1	$+$	0	$-$	1	$+$	$\sqrt{3}$	$+$	f''

singular points: $x^2 - 1 = 0$. vertical asymptotes $x = \pm 1$.

asympote: $y = x$, for $\lim_{x\to\pm\infty}\left(\frac{x^3}{x^2-1} - x\right) = 0$

critical points: $f' = 0$. $x = 0, \pm\sqrt{3}$.

interval of increase: interval that $f' > 0$. $(-\infty, -\sqrt{3})$ or $(\sqrt{3}, \infty)$.

interval of decrease: interval that $f' < 0$. $(-\sqrt{3}, \sqrt{3})$.

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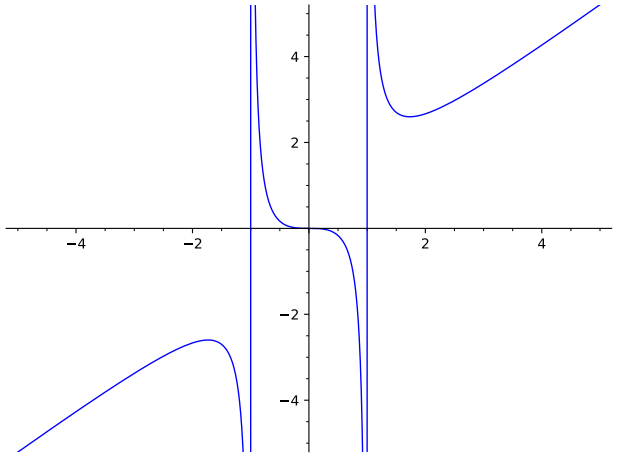

Extreme-Value Problems

Linear Approximations

Taylor Polynomials

Integration by Parts

Integrals of Rational Functions

Inverse Substitutions

Other Methods for Evaluating Integrals

Improper Integrals